

Opportunities in Regenerative Plantations for Kerala

**A Strategic Framework for Policy Makers, Corporate
Leaders, the Startup Ecosystem, and Farming Communities**

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ES | Executive Summary



Kerala's plantation sector — spanning over 7.50 lakh hectares of rubber, tea, coffee, and spices — is the economic backbone of the state's highlands and provides permanent employment to 4.18 lakh workers. Yet the sector stands at a crossroads.

Decades of conventional monoculture farming have degraded soils, eroded biodiversity, and left smallholder farmers chronically vulnerable to price shocks and climate extremes. Shifting temperature and rainfall patterns are expected to significantly impact the plantation sector with up to 45% yield loss expected due to climate change. Kerala's plantation landscape cannot sustainably continue on this trajectory.

At the same time, national and global markets are transforming in ways that create unprecedented opportunity:

- The Central and State government projects such as PM PRANAM Scheme and the Kerala Climate Resilient Agri-Value Chain Modernisation (KERA) project provide the impetus for regenerative agriculture.
- The global regenerative agriculture market reached USD 10–13 billion in 2024, projected to grow at 14–19% CAGR through 2033, driven by corporate net-zero pledges and carbon market growth. Sustainability-marketed commodities command an average of nearly 30% price uplifts premium.
- Certified carbon credits in conservation agriculture can generate additional income of up to ₹2,277 per tonne of carbon.
- The Business Responsibility and Sustainability (BRSR) reporting framework by SEBI, covering top 1000 companies in India, has extended the disclosure framework beyond equity-listed companies to also cover ESG-labelled Debt Securities.
- At the international level, a growing “spaghetti bowl” of regulatory and market-driven frameworks—including the EU Deforestation Regulation (EUDR) and the Corporate Sustainability Due Diligence Directive—is placing increasing pressure on entities to meet higher environmental and human rights

standards. For example, the EUDR, which will be enforced from December 2026, requires exporters of rubber, coffee, and spices to prove deforestation-free status — threatening Kerala's EU access, but rewarding those who prepare early.

In response to these rising global trends, regenerative agriculture offers a practical pathway for compliance and differentiations. Importantly, it is not a foreign concept for Kerala. The state's traditional homestead farming, its shade-grown coffee agroforests in Wayanad, and its centuries-old multi-storeyed spice gardens are all proto-regenerative systems. This white paper argues that Kerala has a rare opportunity to codify, scale, and brand these practices.

Transitioning to regenerative systems — replacing monocultures with polycultures, restoring soil health, integrating shade trees, and recentring community knowledge — is both ecologically sound and economically compelling. However, the barriers are real — chief among them a 2–3 year transition income gap that no smallholder can absorb without patient capital, alongside weak soil diagnostics infrastructure, high certification costs, and legal frameworks designed for monoculture that constrain diversification — but they are addressable with coordinated action.

Making this transition requires a coordinated effort across actors who each have a distinct and irreplaceable role. Estate holders and smallholder farmers must shift land management practices crop by crop with outcomes measured through soil carbon, biodiversity counts, and input reduction. Wholesale buyers and aggregators must build traceability into sourcing, link premium pricing to verified regenerative practice, and help co-develop the market narratives that unlock access to specialty and sustainability-driven buyers. The commodity boards — Rubber Board, Tea Board, Coffee Board, and Spices Board — must set regenerative standards, create compliance readiness pathways aligned with EUDR and other trade requirements, and bring Kerala's story of ecological farming to international markets. Across all crops, the immediate priorities are baseline ecological assessments, agroforestry design support, MRV frameworks to verify outcomes, and certification readiness. No single actor can drive this alone; its success depends on all parts moving in the same direction.

For policy makers, corporate leaders, startups, farming communities, and estate owners alike, the cost of inaction now exceeds the cost of transition. The window is open, and the knowledge to act already exists within Kerala's own landscape.

The Central Argument

Kerala does not need to reinvent regenerative agriculture. It needs to formally recognise, invest in, build on and scale what its farmers already know — and connect that knowledge to the global markets that are ready to pay for it.

This paper is structured around six questions:

- Where is Kerala's plantation economy today, and what good already exists?
- What is regenerative agriculture and how can it be applied to a Kerala plantation context?
- What is the potential of regenerative agriculture — in markets, trade, profitability, and resilience terms?
- Why should each stakeholder group care?
- What are the barriers in adopting regenerative agriculture?
- What preliminary steps should guide action now?

01 | Kerala's Plantation Economy: A Legacy and a Starting Point

Kerala's plantation sector is one of the most concentrated and consequential agricultural economies in India. The state produces around 75% of India's natural rubber (5.90 lakh hectares), 97% of its black pepper, 91.3% of its cardamom (0.69 lakh hectares), 20.69% of its coffee (0.85 lakh hectares), and 18% of its tea (0.36 lakh hectares). The rubber industry contributed approximately 5.4% of Kerala's GSDP in FY24. Important good practices from the Kerala plantation sector are noted below:

- The traditional homestead garden, a multi-storeyed system integrating coconut, arecanut, pepper, fruit trees, and medicinal plants, is a model of regenerative polyculture that predates the term itself.
- Shade-grown coffee agroforests in Wayanad, often intercropped with pepper vines, cardamom, and native forest trees, mirror the agroforestry designs now commanding premium prices in global specialty markets.
- Tribal communities in Wayanad and Idukki have maintained sophisticated knowledge systems around plant use, soil management, and polyculture design that predate modern agronomy.
- The Meenangadi Panchayat in Wayanad has pioneered a carbon neutral village program, with 83.3% of community members (72.4% farmers) surveyed expressing support for agroforestry integration..
- Kerala's cooperative tradition — through institutions like Consumerfed and Marketfed— provides ready-made organisational infrastructure for collective certification.
- Geographical Indication (GI) tags for Wayanad pepper, Idukki cardamom, and Monsooned Malabar coffee provide brand authenticity that most competing origins lack.



The opportunity is to formalise and reward what farmers already know, layer modern science, market access and innovative financing on top, and validate benefits through standardised protocols to create a system where ecological knowledge put in practice becomes a marketable credential, with new revenue streams.

The Stresses Facing the Sector

Despite this foundation, the sector is under compounding structural stress. Most Kerala's plantation farmers are smallholders — 92% of total production and 89% of rubber land is in holdings of less than 2 hectares. These smallholders lack aggregation power, financing access, and market information to respond to price volatility, reduced margins, intensifying competition, or shifting export market requirements.

Rubber prices have been deeply volatile, with a production CAGR of -6.2% recorded during 2009–2015 when prices crashed. Kerala's rubber production stands at 4.91 lakh tonnes as of 2024, a significant decline from a peak of 7.99 lakh tonnes in 2012. Spice production declined from 2.05 lakh tonnes in 2018 to approximately 1.50 lakh tonnes in 2025 — a 26.7% drop. Tea faces labour cost pressure and aging crops and estate infrastructure. Critically, climate change is expected to cause substantial yield losses in all major plantation crops in Kerala. The yield of rubber is expected to reduce by around 25 to 45% by 2050. Coffee and cardamom already experience up to 50% crop loss in periods of unfavourable rainfall extremes, and tea yields are expected to decrease due to rising temperature and deficient rainfall.

02 | Transitioning to Regenerative Systems

What is Regenerative Agriculture?



Regenerative agriculture is an approach to land stewardship that seeks not merely to sustain, but to actively restore and enhance ecological systems. It works with living processes—soil microbiology, water cycles, biodiversity, and climate interactions—to rebuild fertility, increase resilience, and deepen relationships between people and nature. Unlike mainstream industrial agriculture, which often relies on monocultures, external chemical inputs, and extractive practices, regenerative systems emphasize diversity, closed nutrient loops, and minimal disturbance. Plantation agriculture, particularly in colonial contexts, is historically organized around large-scale monocropping for export, often degrading soil, reducing biodiversity, and displacing local knowledge systems. Sustainable agriculture attempts to reduce harm and maintain existing conditions, but regenerative agriculture goes further—it aims to heal damaged ecosystems and improve them over time.

This ethos resonates with principles found in indigenous architecture, where structures are designed as extensions of local ecologies—responsive to climate, materials, and cultural rhythms, rather than imposed as external templates. Similarly, regenerative agriculture is location-based, adaptive, and knowledge-rich.

Regenerative agriculture steers towards cooperation with nature where agriculture, human community and nature cooperate to create an interdependent system. Below are some key considerations to be accounted for while transitioning to regenerative agriculture.

Table 1: Key Steps and Transition Pathways for Regenerative Agriculture

Element	Regenerative Goals	Key Steps for Transition
Soil	Build soil organic matter, eliminate chemical dependency, enhance biological activity. Consider soil as a living system. Rebuild stability and strength to build resilience against erosion and landslides.	• Continuous mulching (coconut husk, leaf litter) • Replace soil fertilizers with natural mixtures and compost • Introduce cover crops (legumes) • Reduce tillage, maintain root systems • Periodic soil testing and carbon tracking
Water	Improve water infiltration, retention, and watershed health Reduced runoff in monsoons, improved groundwater recharge in midlands and high ranges. Circular water systems	• Contour bunding and trenching • Restore ponds, check dams • Increase tree cover for infiltration • Mulching to reduce evaporation • Protect riparian buffers and streams
Biodiversity	Increase on-farm diversity and ecosystem resilience. Shift from monoculture plantations to agroforestry mosaics resembling native ecology Build in systems that integrates and have a positive relationship with native plant and animal systems	• Multi-layer cropping (trees + shrubs + ground crops) • Integrate native species (jackfruit, neem, wild figs) • Create pollinator habitats • Reduce pesticide use; adopt bio-inputs • Maintain uncultivated biodiversity patches
Human & Community	Ensure fairness, dignity, and economic resilience for farmers and workers. Revitalise agricultural work as a livelihood with improving value. Create recognition for indigenous knowledge systems	• Create knowledge flows that link education and agricultural systems. Strengthen farmer cooperatives • Enable local processing and value addition • Revive traditional ecological knowledge • Promote direct market linkages (local + ethical markets)

In the Kerala plantation context—tea, rubber, coffee and spices—this implies transitioning from monoculture estates toward polycultures, integrating shade trees, restoring soil organic matter, and reviving traditional agroforestry practices. It also means re-centering local communities as knowledge holders, transforming plantations from extractive landscapes into regenerative, biodiverse, and socially just ecosystems.

Table 2: Regenerative Transition Pathways by Crop and Farm Scale

Crop	Smallholder Farmers (1–5 acres)	Large Parcel Plantation
Natural Rubber	• Intercrop with native fruit and tubers in early years • Introduce native shade trees along boundaries • Use leaf litter + cow dung-based compost instead of chemical fertilizers • Reduce soil disturbance; allow ground cover vegetation • Integrate small livestock (goats, poultry) for nutrient cycling	• Transition from monoculture to multi-strata agroforestry (rubber + timber + fruit trees) • Replace chemical inputs with bio-inputs at scale • Restore riparian buffers and biodiversity corridors • Phase-wise diversification to avoid income shocks • Worker-led composting and soil regeneration programs
Tea	• Introduce shade trees and windbreaks • Shift to organic pest management • Maintain continuous ground cover to prevent erosion • Small-scale composting using local biomass • Diversify edges with spices (pepper, ginger)	• Reconfigure estates into shaded tea systems • Reduce chemical dependency through integrated pest management (IPM) • Landscape-level water management (contour trenches, ponds) • Convert sections into mixed cropping zones • Train estate workers in regenerative practices
Coffee	• Strengthen traditional shade-grown coffee systems • Increase diversity: pepper vines, fruit trees, medicinal plants • Use on-farm biomass for mulching and soil health • Encourage pollinator habitats • Community seed sharing of native species	• Enhance multi-layered canopy (coffee + native trees + spices) • Systematic soil regeneration plans (cover crops, mulching) • Reduce external inputs through closed-loop nutrient systems • Certification pathways (organic, regenerative) for premium markets • Biodiversity mapping and conservation zones
Spices (Pepper, Cardamom, etc.)	• Grow under mixed tree cover rather than isolated plots • Use traditional pest control (ash, herbal decoctions) • Maintain soil moisture with heavy mulching • Integrate with food crops for resilience • Revive indigenous varieties	• Transition from intensive monocrop spice blocks to forest-like systems • Reintroduce native canopy species for microclimate regulation • Large-scale composting and biomass recycling • Water conservation through shaded systems • Collaborate with local communities for knowledge integration

03 | The Imperative for Change: Why Regenerative, Why Now

Kerala's plantation sector is impacted by multiple stressors. Increasingly erratic rainfall and rising temperatures signalling a challenging future for all the major plantation crops. A problem no doubt exacerbated by widespread monoculture farms which are more vulnerable to crop loss from climate and disease and degrade soil quality and fertility over time, requiring frequent inputs of synthetic fertilizers. Economically, farmers have to tackle competition in the international market with produce from Vietnam and Indonesia, alongside chronic rubber price volatility. The burden falls hardest on smallholders, who own around 99% of rubber and coffee and a majority of cardamom plantations, yet lack collective bargaining power and receive no premium for quality, origin, or sustainability — leaving the bulk of plantation revenue concentrated among processors, traders, and exporters. Overall, Kerala's plantation sector is in decline. The sector reached a peak value of ₹21,000 crore in 2013-14 and has since declined to ₹13,542 crore as of 2020-21. This decline is linked to increased production costs (see table below for cost distribution), specifically labour, which are not matched by increases in commodity prices, necessitating additional revenue. Regenerative agriculture offers a coherent response to all these challenges simultaneously.



Table 3: Cost Structure of Kerala's Plantation Sector

Commodity	Cost of Production (Rs/kg)	Labour (%)	Input Costs (%)	Management (%)	Labour Welfare (%)
Rubber	170	64.63	27.08	5.01	2.64
Cardamon	1082	43.37	52.90	2.29	1.44
Coffee	86	13.65	30.18	19.64	36.53
Tea	190	52.50	34.55	10.16	2.80

Table adapted from GoK-IIM Kozhikode final report on Study on Diversification and Modernization of the Plantation Sector in Kerala, September 2024



There is substantial scientific evidence that regenerative practices improve climate resilience of plantations. Shade-grown coffee agroforestry systems experience 2-4°C lower temperatures, and buffer against climate extremes. For rubber, intercropping with cover crops, legumes, and food crops during the immature phase (for 6-7 years before tree tapping) restore soil microbiome health, reduce chemical inputs and generate additional household income. For tea estates, water harvesting and shade integration addresses both drought and flood extremes. For pepper and cardamom, a multi-storeyed garden model provides built-in climate buffering.

Beyond resilience, agroforestry systems are significantly more effective carbon stores: coffee agroforestry sequesters 70–80 tonnes of carbon per hectare, compared to fewer than 10 tonnes per hectare in monoculture systems.

These environmental gains also translate into economic opportunity. The enhanced carbon sequestration of agroforestry systems opens an additional income stream through carbon credits. Multi-tiered agroforestry models further reduce exposure to single-crop price volatility by generating multiple stacked revenue streams from the same land. Furthermore, regenerative agriculture, when structured through FPOs and collective certification pathways, can direct value towards smallholder farmers in ways that conventional commodity trading fails to do. Certified regenerative produce also commands premium pricing in international markets, providing an additional buffer against commodity price swings. Taken together, the environmental, social, and economic case for transitioning to regenerative plantations is compelling.

A convergence of government incentive programmes, premium markets, emerging finance, and regulatory compliance requirements makes the present moment opportune for this transition. Various central and state government programmes provide financial incentives for sustainable practices which directly support a regenerative transition. Those which are accessible to farmers in Kerala are summarised in the table below. This transition is further supported by a growing premium market, with customers willing to pay an average premium of around



30% for sustainability-marketed products. The global regenerative market is already valued at USD 10–13 billion as of 2024 and is projected to grow at 14–19% CAGR through 2033, driven by corporate net-zero pledges and carbon market growth. Emerging finance options such as India's carbon market framework also promises additional revenue streams for regenerative farms. India's Voluntary Carbon Market (VCM) framework for the agricultural sector was formalised in 2024, and the Carbon Credit Trading Scheme (CCTS) is developing a compliance carbon market to be active from 2026. These carbon markets offer a structural new revenue stream for plantation farmers, complementing global carbon markets such as Verra certified credits which are priced at ₹1,366 to ₹2,277 in over-the-counter transactions. Finally, recent and upcoming regulatory compliance requirements strengthen the case for a regenerative transition. SEBI's BRSR (Business

Responsibility and Sustainability Reporting) framework now requires India's top 1,000 listed companies to disclose supply chain sustainability practices. This includes a new regulation regarding BRSR reporting of ESG debt securities, which regenerative agriculture projects are eligible for under 'social bond' and 'sustainability bond' regulations. This incentivises corporate procurement teams to seek verified sustainable sourcing in their agricultural suppliers. Another key regulation is the EU Deforestation Regulation (EUDR), scheduled to take effect from December 2026 for large operators and June 2027 for small producers. This will directly impact Kerala's plantation exports by requiring proof that products are deforestation-free, reflecting growing consumer demand for environmentally responsible goods in European markets. By proactively meeting these requirements, Kerala's plantation sector could secure a significant first-mover advantage in the EU.

Table 4: Government Schemes Supporting Regenerative Transition in Kerala

Scheme	Level	Focus	Key Financial Support
PKVY	Central	Provide end-to-end support for organic farmers; form organic clusters to create a supply chain	₹31,500/ha over 3 years for States/UTs of which ₹15,000/ha provided to farmers
Subhiksham Surakshitham	Kerala (Central+State)	Natural farming, biodiversity, PGS certification	Initial commitment of ₹44.55 crore. One time farmer's incentive of ₹1600/ha
PM-PRANAM	Central	Reduce chemical fertilizers; reward states for savings	50% of fertilizer subsidy savings as state grant
Organic Farming & GAP	Kerala	Organic certification, safe food, GAP clusters	₹6 crore (2024–25) total allotment
NMNF	Central	Strengthen natural farming practices to promote sustainability, climate resilience and safe food.	₹5000 per ha/year for three years incentive; ₹2,481 crore total outlay
KERA	Kerala + World Bank	Organic & regenerative practices; value chain modernization; certification and branding	₹1821.80 crore loan from World Bank + ₹778.85 crore commitment by Government of Kerala. Financial grants for crop replantation, irrigation and water storage infrastructure, and crop certification

Kerala's plantation sector can no longer afford to delay this transition. The convergence of climate risk, market pressure, and a rare alignment of supportive regulatory frameworks makes the case for regenerative agriculture not merely persuasive, but urgent and critical. The time to act is now.

04 | Why Each Stakeholder Should Care

The regenerative transition is not a single-actor story. Its success depends on a matched set of decisions from policy makers, corporations, startups, farming communities, and estate owners — each of whom has a distinct reason to act, and a distinct cost of inaction.

Table 5: Why Stakeholders Should Act

Stakeholder	Why Act	Cost of Inaction
Policy Makers	EUDR enforcement by Dec 2026 threatens Kerala's EU export access. Climate change reduces yield without resilient practices. Regenerative transition is a rural income and climate resilience agenda simultaneously.	Loss of EU market access for rubber, coffee. Continued climate vulnerability in plantation districts. Missed first-mover advantage as competing states build regenerative frameworks.
Corporate Leaders	SEBI BRSR mandates supply chain sustainability disclosure for top 1,000 listed companies. EUDR compliance requires verified deforestation-free sourcing. Regenerative supply chains reduce Scope 3 emissions and price volatility exposure.	Locked out of EU markets post-December 2026 for non-compliant supply chains. Reputational and legal risk under CSDDD. Loss of supplier base as competitors build certified supply chains first.
Startup Ecosystem	A largely unaddressed market: soil diagnostics, carbon credit aggregation, transition finance products. Anchor customers available through Rubber Board, Spices Board, Coffee Board, and NABARD.	Window for first-mover platform advantage closes as national and international agri-tech players enter. EUDR compliance deadline creates a fixed market timing opportunity.
FPOs & Farmers	Collective certification reduces per-farmer certification costs. Premium pricing creates a floor against commodity price volatility. Carbon revenue adds a structural new income stream from existing land.	Continued commodity price exposure with no premium pathway. Exclusion from EU and North American premium markets as EUDR compliance becomes a market entry requirement. Loss of competitive position as certified FPOs capture premium buyers.
Estate Owners	Carbon revenue from diversified agroforestry. First-mover advantage on Rainforest Alliance and ROC certification before EUDR enforcement. Agri-tourism revenue from regenerative landscapes.	Missed carbon revenue window as compliance market prices rise from 2026. Loss of EU buyer relationships as EUDR enforcement locks out undocumented supply chains. 3-6month certification processing time after transition means action must begin now.

05 | The Potential: Trade, Revenue, and Market Opportunity

The case for regenerative transition is not built on environmental aspiration alone. It is built on hard market dynamics: compliance requirements that will lock non-certified exporters out of premium markets, premium pricing that rewards verified sustainable produce, and new revenue streams from carbon and emerging innovative financing products and structures that did not exist five years ago.

The Market Opportunity

Kerala's four plantation commodities each face distinct global dynamics — and in each case, the regenerative premium pathway is the viable strategic response:

Table 6: The Global Market Opportunity for Regenerative Produce

Commodity	Global Market	Competition	Regenerative Premium Opportunity
Natural Rubber	₹185824 crore (2025); 4.53% CAGR (2026-2034) India is a net importer, importing 5.5 lakh tonnes in 2024-25 compared to an export of 4,839 tonnes.	Thailand (48.5 lakh tons), Indonesia (35.5 lakh tons), Vietnam (11.9 lakh tons), Cote d' Ivoire (1286 MT) — India (9,1 lakh tons)	EUDR-compliant certified rubber; carbon-credited agroforestry; sustainability premium; Sustainable finance
Spices	₹192564 crore (2025) 5.16% CAGR (2026-2034) ₹39994 Indian exports (2024-25) 17% growth in volume from 2023-24	Vietnam, Guatemala, Indonesia — ASTA certified at scale	GI + organic/regenerative + EUDR traceability: up to 50% premium over bulk commodity; Sustainable finance; Carbon credit opportunity
Coffee	₹2271238 crore (2025) 5.4% CAGR (2026-2033) ₹11751 crore (2023-24) Indian exports; increased from ₹6553 crore in 2020-21	Brazil, Vietnam, Colombia, Ethiopia	Shade-grown Western Ghats origin; specialty + EUDR compliance: price premium; Sustainable finance; Carbon credit opportunity
Tea	₹633167 crore (2025) 6.5% CAGR (2026-2033) ₹8417 crore (FY25) Indian export, increased from ₹7524 crore in FY24	China Kenya, Sri Lanka, Vietnam, Indonesia	Orthodox + organic certification: significant premium over CTC commodity grades; Sustainable finance; Carbon credit opportunity

Concrete Examples of Goods and Premiums

- Wayanad shade-grown coffee sold at a premium over commodity Robusta — Kelachandra Coffee's 6,500-acre estates plans to use estate-specific QR-code traceability as a differentiator for European buyers.
- Continental and GIZ are already mapping 5,000 smallholders in Indonesia for EUDR pre-compliance — demonstrating that large tire manufacturers are actively seeking verified sustainable supply chains.
- Organic Nelliampathi tea (Palakkad) demonstrates the premium pathway for Kerala's high-altitude orthodox teas in international markets.
- Carbon sequestration from a well-managed 50-hectare Wayanad coffee estate with native shade trees can generate 70-80 tonnes of carbon sequestration per hectare — similar to that of a tropical forest and potentially valuable in carbon markets.



The Market Is Already Moving

The EU Deforestation Regulation enforces from December 2026. The global regenerative agriculture market reached USD 10–13 billion in 2024. Nestlé alone committed CHF 1.2 billion to regenerative sourcing. India's Carbon Credit Trading Scheme launches in 2026. These are not future scenarios — they are present market realities.

06 | Barriers and Their Real-World Impact



Understanding why good practices have not scaled is as important as understanding what those practices are. The barriers to regenerative adoption in Kerala's plantation sector are multi-layered and mutually reinforcing. Each carries a concrete cost — to farmers, exporters, investors, and the state.

Table 7: Key Barriers to Regenerative Transition

Dimension	Barrier	Impact if Unresolved
Economic	Transition income gap: 2–3 year yield dip; limited patient capital; land tenure insecurity	Smallholder farmers cannot afford the transition. No framework, subsidy, or awareness campaign achieves scale unless this gap is financed. Tribal communities in Wayanad and Idukki are excluded from all long-term investment.
Technological	Limited soil diagnostics-soil microbiology ignored; poor agricultural extension service capability.	Farmers cannot measure or verify the soil health outcomes that certification and carbon markets require. Premium market access and carbon revenue remain inaccessible without this infrastructure. Limited extension service capability hinders dissemination of agricultural technologies and management practices for regenerative transition
Market	Limited long-term contract farming opportunities for premium products.; Certification costs;	Without a price signal, farmers face transition costs with no assured return.
Social	Gender exclusion from FPO governance; caste dynamics; generational migration from farms	Women perform the majority of plantation labour but are systematically excluded from decision-making and formal income recognition. Youth migration hollows out the farm workforce, while indigenous knowledge of polyculture systems is lost with each generation.



A relatively underexamined challenge in transitioning Kerala’s plantation sector toward regenerative models is the existing legal landscape. Several provisions, while not designed to discourage diversification, may have that practical effect.

The Kerala Land Reforms Act (1963) ties plantation status exemptions to monoculture cultivation, creating compliance uncertainty for estates introducing crop diversification. The Plantations Labour Act (1951) similarly defines plantations around specific crops, generating ambiguity around labour regulations for mixed-crop operations.

At the national level, carbon market frameworks are evolving encouragingly through the VCM agriculture framework (2025) and the Carbon Credit Trading Scheme under the Energy Conservation (Amendment) Act 2022. However, gaps remain relevant to Kerala — including the absence of standardised MRV protocols for plantation agroforestry, limited legal clarity around smallholder carbon credit aggregation, and no state-level carbon registry or Kerala-specific additionality baseline.

Aligning these regulatory frameworks with evolving agricultural, environmental, and market priorities would meaningfully strengthen the enabling environment for regenerative transition across the sector.

The Transition Income Problem — The Most Critical Barrier

Regenerative practices typically involve a 2–3 year period during which yields may dip before soil ecology recovers. For a smallholder rubber or spice farmer with no savings and no access to patient capital, this gap is existential. No policy framework achieves scale unless this gap is explicitly financed.

07 | A Multi-Stakeholder Framework for Regenerative Transition

This table outlines a **multi-stakeholder framework for transitioning Kerala's plantation systems toward regenerative agriculture**, grounded in measurable ecological and social outcomes. It clarifies the roles of farmers, estates, market actors, regulatory bodies such as the Tea Board of India, Coffee Board of India, Spices Board of India, and Rubber Board, as well as the enabling role of experts in designing and verifying transitions. Together, it serves as an actionable blueprint for a **five-year, outcome-driven shift from extractive plantation models to regenerative, biodiverse, and economically resilient landscapes**.

Table 8: Multi-Stakeholder Framework for Regenerative Transition

Natural Rubber	
Plantations / Estate Holders	• Role: Transition practices, Cooperate at ecosystemic scales • Outcomes: increased soil carbon, increased plant and tree diversity, decreased chemical inputs • Measurement: Annual soil tests, tree inventory, input logs, biodiversity counts, GIS mapping of agroforestry transition • Opportunity: carbon and biodiversity credit markets, increased climate and financial resilience, long term stability of system
Smallholder Farmers	• Role: Implement regenerative land management practices • Outcomes: Diversified income + improved soil fertility • Measurement: Simple farm diaries, photo documentation, periodic soil testing via FPOs • Opportunity: Carbon and biodiversity credit markets, increased climate and financial resilience
Wholesale Buyers / Aggregators	• Role: Demand traceable, regenerative rubber • Outcomes: End to end traceability, integrating stories into product, diversified participation (Branding) • Measurement: Supplier audits, traceability systems, premium-linked sourcing • Opportunity: Access niche markets and premium sectors
Regulatory Authorities (Rubber Board)	• Role (Rubber Board): Create regenerative standards, facilitate trade compliance readiness, incentivise agroforestry • Outcomes: standardised benchmarks, compliance readiness across supply chain, diversified participation • Measurement: Regional soil + biodiversity monitoring frameworks • Opportunity: First mover advantage, Setting a global standard, maintaining and upscaling the Kerala system

Immediate Steps: Baseline assessments, agroforestry design, carbon + biodiversity tracking systems, MRV (Monitoring, Reporting, Verification) frameworks

Tea	
Plantations / Estate Holders	• Role: Prepare transition pathways at ecosystemic scales. Natural systems management and restoration and regeneration integrated into estate strategy • Outcomes: increased shade cover, biodiversity, and water abundance, decreased erosion, • Measurement: Canopy cover mapping, erosion indicators, water table, pesticide reduction logs • Opportunity: Biodiversity credits and carbon credits as additional and steady sources of income. Global market favour in light of climate and ecological regeneration.
Smallholder Farmers	• Role: Transition to regenerative practices and create cooperative relationships with each other • Outcomes: Reduced input cost, improved resilience • Measurement: Input-output tracking, visual soil health indicators, community biodiversity registers • Opportunity: Cooperative and collaborative models for impact centric product markets. Biodiversity and carbon credits. Community upliftment
Wholesale Buyers / Aggregators	• Role: Build regenerative tea value chains • Outcomes: Bridge between global shift and local value. Access to niche markets and unique positioning. first mover advantage. Branding • Measurement: Certification support, traceability tech, impact reporting to buyers • Opportunity: GI tagging, Branding and creating sourcing systems with integrated regeneration narratives to access niche markets
Regulatory Authorities (Tea Board of India)	• Role: Create strategic shifts with long term planning. Introduce regenerative guidelines, link to export incentives. bringing story to the forefront. • Outcomes: Branding and building unique value with local systems. strengthening positioning in sustainability linked value chain, standardised benchmarks • Measurement: Estate-level sustainability audits • Opportunity: Differentiate Kerala tea as a climate resilient origin, align with global buyer sustainability mandates

Immediate Steps: Transition roadmaps for estates, training modules for workers, erosion & water monitoring systems, certification readiness (ROC, Rainforest Alliance, others)

Coffee	
Plantations / Estate Holders	• Role: Natural systems management and restoration and regeneration integrated into estate strategy • Outcomes: increased multi-layer biodiversity, increased pollinators, stable yields • Measurement: Species counts, yield stability data, soil organic matter tracking • Opportunity: Biodiversity credits and carbon credits as additional and steady sources of income. Global market favour in light of climate and ecological regeneration
Smallholder Farmers	• Role: Transition to regenerative practices and create cooperative relationships with each other • Outcomes: Strengthened agroforestry systems, diversified income • Measurement: Crop diversity logs, participatory biodiversity monitoring • Opportunity: Cooperative and collaborative models for impact centric product markets. Biodiversity and carbon credits. Community upliftment
Wholesale Buyers / Aggregators	• Role: Premium pricing for shade-grown regenerative, and biodiversity friendly coffee • Outcomes: Maintaining traceability, building cooperative relationships between growers and creating systems of aggregation that allow for regenerative value systems, Branding • Measurement: Traceability + impact storytelling (critical for global markets) • Opportunity: GI tagging, Branding and creating sourcing systems with integrated regeneration narratives to access niche markets,
Regulatory Authorities (Coffee Board of India)	• Role: Promote shade-grown and biodiversity standards, support farmer training. Bringing story to the forefront. Support narrative shifts. • Outcomes: established biodiversity standards, increased access to premium markets • Measurement: Landscape-level biodiversity indicators • Opportunity: Leverage existing shade-grown tradition to lead globally on regenerative coffee standards
Immediate Steps: Biodiversity audit frameworks, market linkage strategy (regenerative branding), support for biodiversity credits pilots, Story building and vision	

Spices	
Plantations / Estate Holders	• Role: Natural systems management and restoration and regeneration integrated into estate strategy

	• Outcomes: Forest-like systems restored, increased soil moisture retention, wildlife and biodiversity • Measurement: Canopy density, soil moisture levels, reduction in chemical residues. Increased pollinator presence. • Opportunity: Biodiversity credits and carbon credits as additional and steady sources of income. Increasing the potency and quality of the spices and thereby increased market value
Smallholder Farmers	• Role: Natural diversifying into lesser-known spices, herbs and medicinal plants. Maintaining natural systems. Completely organic and wild systems-based agroforestry at small scale. • Outcomes: Resilient mixed cropping systems. Increased value for lesser indigenous species • Measurement: Yield diversity, soil cover indicators, traditional knowledge documentation • Opportunity: Biodiversity credits and carbon credits as additional and steady sources of income. Capturing the wellness and medicinal markets with indigenous forest species beyond the main spices
Wholesale Buyers / Aggregators	• Role: Build high-value regenerative spice markets • Measurement: Residue testing, traceability, regenerative sourcing claims • Outcomes: Chemical free and high-quality sorting and processing facilities that maintain uniqueness. A stable flow of high-quality products. Increased variety of culinary, medical, and wellness ingredients that can serve a high value market, Branding • Opportunity: GI tagging, Branding and creating sourcing systems with integrated regeneration narratives to access niche markets,
Regulatory Authorities (Spices Board of India)	• Role: Support GI + regenerative branding, export incentives. Research and Training to highlight indigenous medicinal, herbs and spices that are lesser known. Supplying indigenous saplings and seedlings to diversify the plants. Marketing to increase value of Kerala Spices. • Outcomes: Strengthening positioning in sustainability linked value chain, standardised benchmarks • Measurement: Quality + sustainability benchmarks • Opportunity: premium position in market plus emerging market in wellness and health.
Immediate Steps: Design of forest-based spice systems, documentation of indigenous practices, linking biodiversity outcomes to premium markets	

07 | Kerala's Regenerative Transition: A Call for Action



Kerala's plantation sector stands at an inflection point. The climate disruption, shifting markets, and regulatory compliance requirements (unequal distribution of proceeds) bearing down on it are not temporary headwinds — they are structural shifts that will permanently reshape plantation farming. The question is not whether Kerala's landscape will change, but whether that change will be managed on Kerala's terms.

The foundation for transition already exists. The ecological knowledge lives in Kerala's shade gardens, spice agroforests, and homestead farms. The institutional infrastructure exists in Kerala's cooperative tradition, its commodity boards, and its GI tags. The market demand is real and growing. Regulatory support to promote regeneration is fast emerging. What has been missing is the connective tissue: financing mechanisms to bridge the transition income gap, MRV infrastructure to verify ecological outcomes, legal reforms to liberate diversified farming from monoculture-era frameworks, and the institutional will to treat regenerative agriculture as a core economic strategy.

Kerala does not need to adopt foreign frameworks. It needs to formally recognise and scale what its farmers already practise and connect that knowledge to the markets and financing that make it viable. The window is open. What is needed now is coordinated action — from the legislature, commodity boards, corporate leaders, entrepreneurs, and the farming communities whose knowledge has sustained this landscape for generations.

The Kerala Advantage No Other Region Can Replicate

High farmer literacy + cooperative tradition + GI-tagged crops + Western Ghats biodiversity provenance. By 2035, Kerala can be the world's first fully documented, scientifically verified, culturally rooted, and commercially validated model of regenerative plantation agriculture in a tropical biodiversity hotspot.

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Authors



Joseph Martin Chazhoor Francis
Vikhyath Premugh



Nisha Mary Poullose

**International Trade
Consultant**

Tara Francis

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